AI Applications in Research

Concepts and the Use of Claude Code as an Agentic Research Collaborator

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Abstract

This document summarizes the main concepts that have emerged around the use of AI—and coding agents such as **Claude Code** in particular—in empirical and academic research. It synthesizes the practical lessons that applied researchers (notably economist Scott Cunningham in his *Mixtape Substack* "Claude Code" series, and Paul Goldsmith-Pinkham in his NBER talk on AI in empirical research) have been documenting through mid-2026. The organizing theme is a simple asymmetry: agents have collapsed the *marginal cost of producing* research output, but they have *not* lowered the cost of *verifying* that output. Managing that asymmetry is the central skill of AI-assisted research.

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1 The Core Economic Shift

1.1 Falling marginal cost of production

Coding agents (Claude Code, Codex, and similar) let a researcher generate data pipelines, estimators, tables, figures, and draft prose at a speed that was previously impossible. The **marginal cost**

of producing an additional specification, robustness check, or analysis has fallen dramatically, and it now requires comparatively *little* human skill to *drive* that production.

1.2 The verification gap

The complementary insight is that the **cost of verification has not fallen proportionally**. Confirming that a result is correct, reproducible, and faithful to the data still demands substantial human expertise—arguably *more* than before, because the volume of machine-generated output that must be checked has grown. The defining risk of AI-assisted research is therefore an **accumulation of unverified output**: many plausible-looking results, any of which may be subtly or grossly wrong.

What got cheaper	**What did *not***
Writing and running code	Confirming code does what was intended
Producing tables and figures	Tracing each number to its source
Trying many specifications	Judging which specification is valid
Drafting prose and documentation	Domain judgment and interpretation

2 "Prose Is Not a Constraint"

A central lesson is captured in Cunningham's phrase **"prose is not a constraint."** Instructions given to an agent—whether in a `CLAUDE.md` rules file, a system prompt, or an in-line command—are treated by the model as *suggestions*, not enforceable rules.

- An agent may follow a written rule ninety-nine times and silently ignore it the hundredth.
- The tendency to exercise **unilateral discretion** appears to grow as tasks get larger, as the context window fills, or seemingly at random.
- There is no *consequence* to the model for violating an instruction: it cannot be penalized, so restating the rule more forcefully (or with frustration) changes nothing structurally.

Observed failure modes include:

- **Hallucinated quantities**—e.g. inventing a sample size ("494 counties") that no data file supports.
- **Programming "in the air"**—generating a figure or table in an interactive shell without writing the code into the reproducible pipeline.
- **Destructive edits**—half-overwriting a file despite an explicit instruction not to.

3 Tools and Code as Real Constraints

If prose does not bind, what does? **Tools bind.** The environment, software packages, and version control impose constraints the agent cannot talk its way around.

3.1 Packages that enforce methodology

A vivid example: the `did` package in R (Callaway and Sant'Anna's estimator for staggered treatment adoption) *requires* the existence of not-yet-treated or never-treated units and will *error out*

otherwise. A plain two-way fixed-effects regression via `lm` or `fixest` will happily run on a fully-treated panel and return a misleading number. Choosing a package that *refuses invalid designs* converts a methodological rule into a hard constraint—something prose alone could never achieve.

3.2 Version control as a safety net

Git turns dangerous, silent actions into visible, reversible ones:

- A fabricated "+494" shows up as a green added line in a diff you must *approve* before it lands—so it gets caught.
- A clobbered file appears as red deleted lines you can *reject*.
- A lost draft is recoverable with a single `git checkout`.

The general principle: **route agent output through mechanisms that make errors surface and mistakes recoverable** rather than trusting the agent to obey.

4 Workflow Principles for AI-Assisted Research

1. **Everything traces to a file.** Every reported number, table, and figure must be produced by code committed to a reproducible pipeline and traceable to a local data file—never generated "in the air."
2. **Reproducible pipelines over ad-hoc commands.** Prefer scripted, re-runnable pipelines so that a change upstream (e.g. the analytical sample moving from N to $N + k$ observations) propagates and is detected rather than hidden.
3. **Constrain with tools, not words.** Encode methodological commitments in package choices, schemas, tests, and type systems that *fail loudly* on violation.
4. **Diff-review every change.** Treat the agent like a contributor whose pull requests you review line-by-line; approve deliberately.
5. **Guard against context drift.** Break long projects into scoped tasks; compliance degrades as context grows, so smaller contexts are safer.
6. **Verification is the human's job.** Human skill migrates from *writing* code to *auditing* results—designing the checks, spotting the implausible, and owning correctness.
7. **Manage deprecation and "the canon."** Keep a curated set of trusted, current methods and packages so agents do not reintroduce stale or deprecated code.

5 Using the Agent as a Thinking Partner

Beyond code execution, Claude Code is used as a **reasoning and reflection tool**:

- **Blind-spot discovery.** Cunningham applies the *Johari Window* (the 2×2 of what is known/unknown to self and others) with the agent to surface unexamined assumptions in a research design—using the model's broad exposure to the project to articulate problems the researcher cannot see.
- **Adversarial critique.** Having the agent read an external expert's slides (e.g. Goldsmith-Pinkham's NBER talk) and use them as a structured critique of one's own project.

- **Documentation and exposition.** Turning working sessions, derivations, and results into readable write-ups and teaching material.

6 Broader Applications of AI in Research

The Claude Code case sits within a wider set of AI-in-research applications:

- **Literature synthesis and review**—summarizing, comparing, and situating prior work.
- **Data cleaning and feature engineering**—wrangling messy real-world data into analysis-ready form.
- **Estimation and simulation at scale**—running many designs, Monte Carlo experiments, or semi-autonomous agent fleets.
- **Code translation and modernization**—porting analyses between languages (e.g. Stata → R → Python).
- **Drafting and communication**—papers, slides, referee reports, and newsletters.

7 Takeaways

1. AI agents make *producing* research nearly free but leave *verifying* it expensive; the gap between the two is where errors live.
2. **Prose is not a constraint**—written rules are advisory, so do not rely on instructions alone for correctness.
3. **Tools, packages, and version control *are* constraints**—design the environment so that invalid work fails loudly and mistakes are reversible.
4. The researcher's comparative advantage shifts to **judgment, design, and verification**: deciding what is valid, and proving that it is.

Primary sources: Scott Cunningham, *Scott's Mixtape Substack*, "Claude Code" essay series, 2026 (https://causalinf.substack.com/s/claude-code); Paul Goldsmith-Pinkham, NBER talk on AI in empirical research, 2026. This brief is a synthesis prepared for internal reference.